DATA DICTIONARY

Overview

The data comes from a soft-drink production line: 38 production batches over four days (29, 30, 31 Aug and 2 Sep 2024), with 61 downtime records caused by 12 possible factors. The model has four tables: two fact tables (Line productivity, Line downtime) and two dimension tables (Products, Downtime factors).

TABLE	FIELD	DESCRIPTION	KEY TYPE	PRIMARY KEY	SAMPLE VALUE
Line productivity	*Fact table: one row for each batch produced*				
	Date	Date the batch was produced	-		8/29/2024
	Product	ID of the product produced in the batch	FK		OR-600
	Batch	Unique ID of the batch produced	PK	Batch	422111
	Operator	Production line operator in charge of the batch	-		Mac
	Start Date Time	Date and time the batch production started	-		8/29/2024 11:50:00 AM
	End Date Time	Date and time the batch production ended	-		8/29/2024 2:05:00 PM
	Production Duration	Total time, in minutes, taken to produce a single batch	-		135 min
Products	*Dimension table: details of each product*				
	Product	Unique product ID	PK	Product	OR-600
	Flavor	Soda flavor of the product	-		Orange
	Size	Product size (volume)	-		600 ml
	Min batch time	Minimum time required to produce a batch (with no downtime)	-		60 min
Line downtime	*Fact table: downtime (in minutes) by factor for each batch*				
	Batch	Unique ID of the batch produced	FK	Composite key (Batch, FactorID)	422111
	FactorID	Unique ID of each downtime factor	FK		2
	DownTime Duration	Downtime minutes caused by the factor in the batch	-		60 min
Downtime factors	*Dimension table: details of each downtime factor*				
	Factor	Unique ID of each downtime factor	PK	Factor	2
	Description	Downtime factor description	-		Batch change
	Operator Error	Is this due to operator (human) error? (Yes / No)	-		Yes